Simon Fraser University
Department of Statistics and Actuarial Science

Suggested courses that may be used towards the A.Stat. designation
2019

Module	**Accredited Course**
Mathematics Modules	
1. Calculus I	MATH 150 – Calculus I with review MATH 151 – Calculus I; **or** MATH 154 – Calculus I for the Bio Sci; **or** MATH 157 – Calculus I for the Social Sci
2. Calculus II	MATH 152 – Calculus II; **or** MATH 155 – Calculus II for the Bio Sci,; **or** MATH 158 – Calculus II for the Social Sci
3. Linear Algebra	MATH 232 – Applied Linear Algebra **or** MATH 240 – Algebra I – Linear Algebra
Statistics and probability modules	
4. Mathematical statistics	STAT 330 – Introduction to Mathematical Statistics **or** STAT 450 – Statistical Theory; **or** STAT 830 – Statistical Theory I (graduate) **or** STAT 831 – Statistical Theory II (graduate)
5. Linear Regression	STAT 350 – Linear Models in Applied Statistics STAT 850 – Linear Models and Applications
6. Design of Experiment	STAT 430 – Experimental Design **or** STAT 850 – Linear Models

7. Survey Sampling	STAT 410 – Sample Surveys
8. Stat Elective	Select an additional three courses.
9. Stat Elective	
10. Stat Elective	STAT 380 – Stochastic Processes; STAT 390 – Special Topics, STAT 440 - Learning from Big Data STAT 452 - Statistical Learning and Prediction STAT 475 – Applied Discrete Data Analysis, STAT 460 – Bayesian Statistics STAT 495 – Special Topics, STAT 445 – Applied Multivar Analysis STAT 485 – Applied Time Series STAT 851 – General Linear Models STAT 855 – Lifetime Data Analysis, STAT 832 – Applied Probability Models, STAT 890 – Selected Topics, STAT 830 – Statistical Theory I STAT 840 – Stat Genetics STAT 841 – Advanced Design of Experiments STAT 843 – Functional Data Analysis STAT 850 – Linear Models STAT 852 – Modern Methods in Statistics STAT 853 – Statistical Computing STAT 854 – Biomedical Studies

	STAT 856 – Longitudinal Methods STAT 857 – Space-Time Models ACMA (Actuarial Mathematics) courses at 300+ level
Computer Skills	
11. Computer skills 1	CMPT 125 - Introduction to Computing Science and Programming II, **or** CMPT 129 - Introduction to Computing Science and Programming for Mathematics and Statistics
12. Computer skills 2	STAT 260/261 - Introductory R for Data Science, **and** STAT 342 - Introduction to Statistical Computing and Exploratory Data Analysis - SAS
Communication Skills	
13. Communication skills	Any Writing Intensive course offered at SFU at 300+ level. Examples: STAT 300W – Statistics Communication, **or** STAT 811 – Statistical Consulting I, **and** STAT 812 – Statistical Consulting II
Substantive Area	
14. Course 1	Three courses at 300+ level from one area other than STAT, MATH, **or** MACM.
15. Course 2	
16. Course 3	

Approved by the Accreditation Committee 2019-05-26. Updated 2021-06-07.